Chapter 1 Review Problem Set

Probability Theory — the Foundation of Random Signal Analysis

21 questions · 7 days · 3 questions per day

How to use this set

- **One question per page.** The ruled area below each question is deliberately left blank for handwritten working. There are no solutions in this booklet.
- **Source tag.** The bracketed line before every question gives where the question comes from: a past exam paper, a textbook exercise, or a formula-recall prompt.
- **Suggested pace.** 3 questions per day, about 20–30 minutes each. Do the formula-recall questions from memory first, then check against your notes.
- **Ordering follows exam frequency**, not textbook order. The topics are ranked by the marks they actually carried across the four past papers.

Where this set comes from

Every question is mapped to the exam-frequency analysis of four past papers (2022–23, 2023–24, 2024–25, 2025–26). Across those papers Chapter 1 carried **29 / 33 / 24 / 49 marks** — an average of **33.8 marks per paper**, and two papers devoted a whole 25-mark question to it. The plan below puts the most time on the highest-yield topics.

Chapter 1 topics covered, by exam frequency tier

Tier	§	Topic	Marks	Hits	Questions
S	1.4.2	Moments and joint moments	37	3/4	Q1, Q2, Q3, Q4, Q7, Q8, Q18
S	1.6	Typical distributions	25	3/4	Q13, Q15, Q16, Q17, Q18
A	1.5.1	Characteristic function (one-dimensional)	18	2/4	Q10, Q11, Q12
A	1.2.1	Multivariate random variables	17	2/4	Q7, Q8, Q9, Q17, Q18
A	1.4.3	Uncorrelatedness and orthogonality	14	1/4	Q5, Q6
B	1.3.1	Function of one random variable	10	1/4	Q14
B	1.4.1	Expectation	8	1/4	Q1, Q2, Q3
B	1.1.2	Random variables	6	1/4	Q8
B	1.2.3	Independence	0	4/4	Q9, Q16
C	1.1.1	Total probability and Bayes' rule	0	0/4	Q19
C	1.3.2	Function of two random variables	0	0/4	Q7, Q16
C	1.4.4	Conditional expectation	0	0/4	Q20
C	1.5.2	Joint characteristic function	0	0/4	Q21

The Seven-Day Plan

Day	Q	Sources	Focus and target skills
1	1, 2, 3	Formula recall (§1.4.1–§1.4.2); Textbook Ex. 1.21; Exam 2023–24 Q3(a)–(c)	Expectation, variance and covariance. Write the definitions from memory, then manipulate E, D and Cov of linear combinations, then normalise a joint density and find E and D from it.
2	4, 5, 6	Exam 2023–24 Q3(d); Textbook Ex. 1.22; Exam 2024–25 Q1(b)	Joint moments and the independence triangle. Extract m_{XY} and ρ_{XY} from a density, then separate the three notions *orthogonal*, *uncorrelated* and *independent* on two contrasting examples.
3	7, 8, 9	Textbook Ex. 1.14; Exam 2025–26 Q1(a)–(c); Exam 2025–26 Q1(d)	Joint distributions. Joint law, marginals and the law of $Z = XY$ for discrete variables; probability densities built from a random experiment; two-dimensional densities and covariance.
4	10, 11, 12	Formula recall (§1.5.1); Exam 2022–23 Q1(b); Textbook Ex. 1.29	Characteristic functions. Definition, Fourier pair, uniqueness and the three properties; then CF of a shifted uniform, and recovering a density and its moments from a CF.
5	13, 14, 15	Formula recall (§1.6); Textbook Ex. 1.17; Exam 2023–24 Q2(c)	Typical distributions and Gaussian transformations. Reproduce six distributions from memory; density of a non-monotone function including its impulse; Gaussianity of a filtered noise difference.
6	16, 17, 18	Textbook Ex. 1.16; Formula recall (§1.2.1, §1.6); Exam 2025–26 Q4(b)–(c)	Transformations and the bivariate Gaussian density. Joint density of $U = X+Y$, $V = X-Y$; the general bivariate Gaussian density and its two-time form; in-phase and quadrature components.
7	19, 20, 21	Textbook Ex. 1.6; Textbook Ex. 1.24; Textbook Ex. 1.33	The remaining topics. Total probability and Bayes' rule, conditional expectation and the tower property, and marginal characteristic functions from a joint CF.

Notes on scope.

- §1.4.3 (uncorrelatedness and orthogonality) carried 14 marks in the past papers but is *not* drilled by the current homework. Q5 and Q6 are included specifically to close that gap.
- §1.1.1, §1.3.2, §1.4.4 and §1.5.2 were never examined directly in the four papers, but every one of them is a *compulsory* homework exercise, so Day 7 covers them for the coursework mark.
- §1.4.5 (Chebyshev and Cauchy–Schwarz inequalities) and §1.3.3 (Rayleigh and Rice distributions) are deliberately omitted: they took 0 marks in all four papers and are not set as homework.

Question 1

Day 1 · Expectation, Variance and Covariance

(Formula recall — §1.4.1 and §1.4.2)

For two random variables X and Y, write down the definition of each of the following quantities. Give both the general definition and the form used when a probability density function is available.

(a) the n-th moment about the origin, m_n;

(b) the n-th central moment, μ_n;

(c) the mean-square value and the root-mean-square value;

(d) the variance $D(X)$ and the standard deviation σ_X;

(e) the joint moment m_{kl} about the origin;

(f) the covariance $\operatorname{Cov}(X, Y)$;

(g) the correlation coefficient ρ_{XY}.

Finally, **derive** the identity

$$D(aX + bY) = a^2 D(X) + b^2 D(Y) + 2ab \operatorname{Cov}(X, Y).$$

Working space

Question 2

Day 1 · Expectation, Variance and Covariance

(Textbook Exercise 1.21)

For the random variables X and Y it is known that

$$EX = 1, \qquad EY = 3, \qquad D(X) = 4, \qquad D(Y) = 16, \qquad \rho_{XY} = 0.5.$$

Let

$$U = 3X + Y, \qquad V = X - 2Y.$$

Find EU, EV, $D(U)$, $D(V)$ and $\mathrm{Cov}(U, V)$.

Working space

Question 3

Day 1 · Expectation, Variance and Covariance

(Final examination 2023–24, Q3(a)–(c))

The joint probability density function of the random variables X and Y is

$$f_{XY}(x,y) = A\sin(x+y), \qquad 0 \le x \le \frac{\pi}{2}, \qquad 0 \le y \le \frac{\pi}{2}.$$

Find

(a) the coefficient A;

(b) the expectations $E(X)$ and $E(Y)$;

(c) the variances $D(X)$ and $D(Y)$.

Working space

Question 4

Day 2 · Joint Moments, Correlation and Independence

(Final examination 2023–24, Q3(d))

For the same joint probability density function as in Question 3,

$$f_{XY}(x,y) = A\sin(x+y), \qquad 0 \le x \le \frac{\pi}{2}, \qquad 0 \le y \le \frac{\pi}{2},$$

find

(a) the joint moment $m_{XY} = E(XY)$;

(b) the correlation coefficient ρ_{XY}.

Hence state, with justification, whether X and Y are uncorrelated, whether they are orthogonal, and whether they are independent.

Working space

Question 5

Day 2 · Joint Moments, Correlation and Independence

(Textbook Exercise 1.22)

The joint probability density function of the random variables X and Y is

$$f_{XY}(x,y) = \begin{cases} \dfrac{xy}{9}, & 0 \le x < 2, \quad 0 \le y < 3, \\ 0, & \text{otherwise.} \end{cases}$$

Determine whether X and Y are **orthogonal**, whether they are **uncorrelated**, and whether they are **independent**. Justify each answer.

Working space

Question 6

Day 2 · Joint Moments, Correlation and Independence

(Final examination 2024–25, Q1(b))

Two random signals are given by

$$X[n] = \sin(2\pi n + \phi), \qquad Y[n] = \cos(2\pi n + \phi),$$

where $\phi \sim U(0, \pi)$ and n is an integer. Calculate and discuss the **orthogonality**, the **uncorrelatedness** and the **statistical independence** of the two random signals. Justify your answer.

Hint: The deterministic identities for integer n simplify both signals a great deal before any averaging is done. Remember that orthogonality and uncorrelatedness are two different conditions, and that neither of them implies independence in general.

Working space

Question 7

Day 3 · Joint Distributions of Several Random Variables

(Textbook Exercise 1.14)

The discrete random variables X and Y have the following joint probability mass function.

$X \setminus Y$	-1	0	1
0	0.07	0.18	0.15
1	0.08	0.32	0.20

Find

(1) the joint distribution function and the joint probability density function of X and Y;

(2) the marginal probability mass functions of X and of Y;

(3) the probability mass function of $Z = XY$;

(4) the correlation coefficient of X and Y.

Working space

Question 8

Day 3 · Joint Distributions of Several Random Variables

(Final examination 2025–26, Q1(a)–(c))

A continuous-time random process is defined by throwing a coin once:

$$X(t) = \begin{cases} \sin(2\pi t), & \text{if the head is obtained,} \\ t^2, & \text{if the tail is obtained.} \end{cases}$$

The probabilities of obtaining a head and a tail are identical. Find

(a) the one-dimensional probability density function $f_X(x, 2)$;

(b) the two-dimensional probability density function $f_X(x_1, x_2; 1, 2)$;

(c) the covariance $C_X(t_1, t_2)$.

Hint: Because the process takes only two possible waveforms, every probability density here is a sum of Dirac impulses. Write the general expression in t_1 and t_2 first, and substitute the numbers only at the end.

Working space

Question 9

Day 3 · Joint Distributions of Several Random Variables

(Final examination 2025–26, Q1(d))

A discrete-time random process is defined by independently throwing a coin repeatedly:

$$Y[n] = \begin{cases} \sin(2\pi n), & \text{if the head is obtained in the } n\text{-th throw,} \\ n^2, & \text{if the tail is obtained in the } n\text{-th throw.} \end{cases}$$

In every throw the probabilities of a head and a tail are identical. Find the two-dimensional probability density function $f_Y(y_1, y_2; 1, 2)$.

Working space

Question 10

Day 4 · Characteristic Functions

(Formula recall — §1.5.1)

Let X be a random variable with probability density function $f_X(x)$ and characteristic function $\phi_X(v)$.

(a) Write the definition of $\phi_X(v)$ as an expectation, and also as an integral.

(b) State the Fourier transform pair that relates $\phi_X(v)$ and $f_X(x)$, and state the uniqueness theorem.

(c) State Property 1: the characteristic function of a sum of independent random variables.

(d) State Property 2: the characteristic function of $Y = aX + b$.

(e) State Property 3: how the moments of X are obtained from the derivatives of $\phi_X(v)$ at the origin. Write the general formula for $E[X^k]$.

Working space

Question 11

Day 4 · Characteristic Functions

(Final examination 2022–23, Q1(b))

A random signal is given by

$$X(t) = U + t, \qquad t \in [-2, 2],$$

where U is a random variable uniformly distributed on $[0, \pi]$. Find the characteristic function of $X(t)$.

Hint: Find the characteristic function of U first, then use the property of Question 10(d). The result should depend on both v and t.

Working space

Question 12

Day 4 · Characteristic Functions

(Textbook Exercise 1.29)

For each of the following characteristic functions of a random variable X, find its probability density function, its mean, its mean-square value and its variance.

(a) $\phi(v) = 0.2 + 0.3e^{j2v} + 0.2e^{j4v} + 0.2e^{-j2v} + 0.1e^{-j4v}$

(b) $\phi(v) = 0.3e^{jv} + 0.7e^{-jv}$

(c) $\phi(v) = \dfrac{4}{4 - jv}$

(d) $\phi(v) = \dfrac{\sin 5v}{5v}$

Working space

Question 13

Day 5 · Typical Distributions and Gaussian Transformations

(Formula recall · §1.6)

From memory, reproduce the following table for six typical distributions. For each one give the probability density function (or the probability mass function), the characteristic function, the mean and the variance.

	Distribution	Density / mass function	Characteristic function	Mean and variance
(a)	Uniform $U(a, b)$			
(b)	Exponential, parameter λ			
(c)	Gaussian $N(\mu, \sigma^2)$			
(d)	0–1 distribution, parameter p			
(e)	Binomial $B(n, p)$			
(f)	Poisson $P(\lambda)$			

Working space

Question 14

Day 5 · Typical Distributions and Gaussian Transformations

(Textbook Exercise 1.17)

The output Y of a half-wave rectifier is related to its input X by the model

$$Y = g(X) = \begin{cases} X, & X \geq 0, \\ 0, & X < 0. \end{cases}$$

The input random variable X has probability density function $f_X(x)$ and distribution function $F_X(x)$. Find the probability density function $f_Y(y)$ of the output.

Hint: g is not one-to-one, and the whole negative half-line is mapped to the single value 0. Handle the continuous part and the discrete value separately, and expect a Dirac impulse.

Working space

Question 15

Day 5 · Typical Distributions and Gaussian Transformations

(Final examination 2023–24, Q2(c))

A zero-mean white noise $W(t)$ with power spectral density $N_0/2$ is applied to a linear time-invariant system whose impulse response is

$$h(t) = e^{-t}u(t).$$

Let $X(t)$ be the output of that system, and form a second signal

$$Y(t) = X(t) - X(t-T), \qquad T > 0.$$

You may use the result

$$R_X(\tau) = \frac{N_0}{4}e^{-|\tau|}.$$

Determine whether or not $Y(t)$ follows a Gaussian distribution. Justify your answer, and give its one-dimensional probability density function.

Hint: $Y(t)$ is a linear functional of a Gaussian process. Its mean and its variance are all you need; the variance follows from $R_Y(0) = 2R_X(0) - 2R_X(T)$.

Working space

Question 16

Day 6 · Transformations and the Bivariate Gaussian Density

(Textbook Exercise 1.16)

Let $X \sim N(0,1)$ and $Y \sim N(0,1)$ be independent random variables. Define

$$U = X + Y, \qquad V = X - Y.$$

(1) Find the joint probability density function $f_{UV}(u,v)$ of the pair (U,V).

(2) Determine whether U and V are independent.

Working space

Question 17

Day 6 · Transformations and the Bivariate Gaussian Density

(Formula recall · §1.2.1 and §1.6)

(a) Write the general bivariate Gaussian joint probability density function $f_{XY}(x, y)$ for means μ_X, μ_Y, variances σ_X^2, σ_Y^2 and correlation coefficient ρ.

(b) State the condition on ρ under which X and Y are independent, and explain why the condition works.

(c) Give the special case $\sigma_X = \sigma_Y = \sigma$ and $\rho = 0$.

(d) For a Gaussian random process $X(t)$, explain how the two-dimensional density $f_X(x_1, x_2; t_1, t_2)$ is obtained. State exactly which two quantities of the process determine it completely.

Working space

Question 18

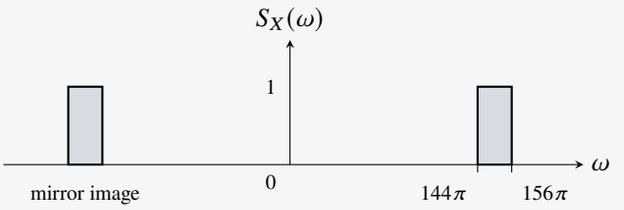
Day 6 · Transformations and the Bivariate Gaussian Density

(Final examination 2025–26, Q4(b)–(c))

A zero-mean stationary Gaussian signal is given by

$$X(t) = i(t)\cos\omega_0 t - q(t)\sin\omega_0 t, \qquad \omega_0 = 148\pi,$$

where $i(t)$ and $q(t)$ are the in-phase and quadrature components. Its power spectral density $S_X(\omega)$ equals 1 for $144\pi \le |\omega| \le 156\pi$ and equals 0 elsewhere.



Find

(a) the joint probability density function $f_{iq}(i, q; t, t)$;

(b) the joint probability density function $f_{iq}\left(i, q; t, t + \frac{1}{12}\right)$.

Working space

Question 19

Day 7 · The Remaining Chapter 1 Topics

(Textbook Exercise 1.6)

There are four batches of components. The first batch contains 2 000 components, of which 5% are defective. The second batch contains 500 components, of which 40% are defective. The third and fourth batches each contain 1 000 components, with approximately 10% defective. A batch is selected at random, and then a component is drawn at random from that batch. Find

(1) the probability that the selected component is defective;

(2) the probability that the component came from the second batch, given that it is defective.

Working space

Question 20

Day 7 · The Remaining Chapter 1 Topics

(Textbook Exercise 1.24)

The random variable X is uniformly distributed on $[0, a]$. Given X, the random variable Y is uniformly distributed on $[X, a]$. Find

(1) $E(Y \mid X)$, for $0 \le X \le a$;

(2) EY.

Hint: For part (2) you do not need to find the density of Y. Use the tower property of conditional expectation.

Working space

Question 21

Day 7 · The Remaining Chapter 1 Topics

(Textbook Exercise 1.33)

The joint characteristic function of the random variables X and Y is

$$\phi_{XY}(u, v) = \frac{6}{6 - 2ju - 3jv - uv}.$$

Find

(1) the characteristic function of X;

(2) the expectation and the variance of Y.

Hint: Both answers come from the same idea: set one of the two variables to zero to obtain the marginal characteristic function, then recognise which distribution it belongs to.

Working space